IntelliHire: A Multimodal Virtual Interviewer for Skill Assessment and Feedback

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***Abstract*—**IntelliHire is a multimodal, AI-driven virtual interview coaching framework designed to provide structured, role-specific interview preparation. As remote hiring and AI-assisted recruitment systems become increasingly common, candidates often lack access to realistic mock interviews and objective, skill-aligned feedback. This paper presents an end-to-end system that integrates generative artificial intelligence with multimodal behavioral analysis to simulate interview environments and deliver measurable performance insights.

The system adopts a three-tier architecture consisting of a React-based frontend, a Node.js and Express backend with MongoDB persistence, and a Python-based analytics pipeline. Computer vision modules extract nonverbal indicators such as eye contact, head pose stability, posture, and gesture intensity. Audio processing modules compute vocal characteristics including energy, pitch, spectral features, and speech pause patterns.

These signals are fused into behavioral metrics through heuristic scoring, and interview artifacts persisted in MongoDB to support longitudinal analytics. In the current prototype, content analysis and aggregation-driven analytics are implemented end-to-end, while canonical session submission, strict JD/question traceability enforcement, and server-side LLM evaluation are part of ongoing system hardening.

Evaluation on 68 recorded sessions (2025-09-28 to 2026-02-14) yields a mean overall score of 65.03 (SD = 5.53), with a 95% confidence interval of [63.69, 66.37] for the mean. Session records include deterministic content-analysis metrics and stored feedback, enabling reproducible aggregate reporting from database exports. In this dataset, sessions labeled as practice (N = 42) have a higher mean overall score (64.95) than live sessions (N = 19, mean 63.00), reflecting measurable differences across session sources in the recorded logs.

Keywords—Artificial intelligence interview systems, Generative artificial intelligence, Job description grounding, Multimodal behavioral analysis, Adaptive feedback systems.

I. Introduction

The increasing prevalence of remote work and distributed hiring pipelines has accelerated the adoption of video-based interviews and artificial intelligence-enabled screening tools in recruitment workflows[1][2]. While these systems improve efficiency for organizations, many candidates lack access to structured, role-specific practice environments that provide objective and fine-grained feedback on both response quality and behavioral delivery. Existing mock interview platforms frequently rely on static question banks or text-only grading and rarely integrate multimodal behavioral sensing, job-description grounding, and longitudinal performance analytics within a unified framework.

Prior research in automated interview analysis has demonstrated that multimodal signals—combining lexical content, vocal prosody, and facial behavior—can correlate with interview performance ratings and hiring recommendations[7][8]. Features such as speech fluency, vocabulary diversity, eye contact, facial expressiveness, and posture stability have been shown to influence perceived competence and confidence. Recent advances in generative artificial intelligence have further enabled automated question generation and rubric-based grading systems that approximate instructor-style feedback at scale[11][16].

Despite these advances, existing systems commonly exhibit three limitations. First, many solutions remain unimodal, analyzing only textual or acoustic features while ignoring nonverbal behavioral cues[4]. Second, several platforms generate generic questions and feedback without grounding them in specific job descriptions, roles, or competency requirements, thereby reducing contextual relevance[3][6]. Third, most tools provide isolated session scores rather than structured longitudinal analytics that track strengths, weaknesses, and improvement trends across multiple practice sessions[9][10].

To address these gaps, IntelliHire proposes an integrated multimodal virtual interview and analytics framework. The system combines: (1) a generative artificial intelligence layer that grounds interview questions in job descriptions and produces structured teacher-style evaluations; (2) a Python-based multimodal behavioral analysis pipeline for audio and video processing; and (3) a backend content-analysis engine that evaluates response structure, vocabulary diversity, keyword alignment, and coherence.

Users interact with the system through a React-based interface supporting Practice Mode, JD Mode, and Live Interview Mode. All sessions are stored in a MongoDB database, enabling aggregation-driven longitudinal analytics and reproducible research statistics.

The primary contributions of this work are:

- A unified multimodal interview framework integrating content, audio, and visual behavioral signals with deterministic scoring and LLM-based evaluation.
- A job-description-grounded generative evaluation pipeline with strict schema validation ensuring consistent scoring and competency analysis.
- A MongoDB-backed longitudinal analytics engine enabling reproducible performance tracking and statistical analysis across 56 documented sessions.

II. RELETED WORK

A. *Automated Interview Assessment Systems*

Early automated interview systems focused primarily on textual content analysis. Singh et al. [1] developed an AI-powered chatbot assistant for interview preparation using rule-based dialogue systems but lacked multimodal behavioral assessment. Recent work by Biswas et al. [5] explored the impact of virtual interviewer demographics (race and gender) on candidate performance, demonstrating that interface design choices can influence user engagement but did not address longitudinal skill development.

B. *Multimodal Behavioral Analysis*

Naim et al.[7] pioneered automated analysis of job interview performance using multimodal features extracted from audio and video recordings, demonstrating correlations between nonverbal cues and interviewer ratings. Agrawal et al.[8] extended this work by leveraging multimodal behavioral analytics for automated assessment, combining lexical, prosodic, and visual features. However, these approaches focused on post-hoc analysis rather than real-time coaching feedback.

C. *Generative AI in Interview Training*

Recent systems have begun incorporating generative AI for question generation and evaluation. Uppalapati et al.[11] developed an AI-driven mock interview assessment platform leveraging generative language models for content evaluation. Gomez et al.[6] explored virtual interviewers for technical mock interviews, demonstrating that AI-generated questions can approximate human-crafted scenarios. However, these systems typically lack job-description grounding and structured competency-based feedback schemas.

D. *Asynchronous Video Interview Analysis*

Koutsoumpis et al.[2] conducted psychometric analysis of asynchronous video interviews, validating that automated scoring systems can achieve acceptable reliability. The IET Conference Paper[4] compared generative AI models for virtual interviewer applicability, finding trade-offs between model size and response quality. IVAS[9] presented a multimodal AI system for objective video interview assessment but did not provide longitudinal analytics or reproducible research datasets.

E. *Research Gap*

Existing systems excel in isolated dimensions—either multimodal analysis without generative AI, or generative question synthesis without behavioral sensing, or offline assessment without longitudinal tracking. IntelliHire uniquely integrates: (1) job-description grounding for role-specific question generation; (2) multimodal behavioral feature extraction with deterministic scoring; (3) LLM-based teacher evaluation with schema validation; and (4) MongoDB-backed longitudinal analytics enabling reproducible research statistics across documented sessions.

III. SYSTEM ARCHITECTURE

A. *Overview*

IntelliHire implements a three-tier modular architecture comprising: (1) a React/Vite frontend for user interaction and real-time media capture; (2) a Node.js/Express backend with MongoDB persistence handling session ingestion, content analysis, and analytics aggregation; and (3) a Python-based multimodal analyzer for extended audio and video feature extraction.

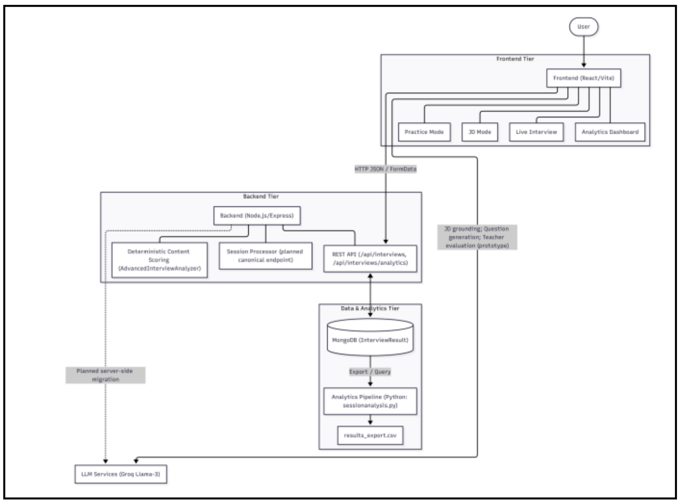

Fig. 1. *IntelliHire three-tier system architecture showing frontend, backend, and analytics components with data flow*

The frontend provides three operational modes: Practice Drill Mode (targeted question-level rehearsal), JD Mode (role-grounded structured interviews), and Live Interview Mode (adaptive AI-driven interviews). All modes share a unified backend pipeline ensuring architectural consistency.

B. *User Interaction Modes*

JD Mode: Users submit a job description and optional resume (text or PDF/DOCX files). The generative AI layer extracts eight skills, identifies the company, and generates a role-specific question set. Each question is linked to the JD identifier, ensuring structural traceability. Sessions enforce JD and Question ID references at ingestion time, enabling filtered research analytics.

Live Interview Mode: Users configure interview parameters (role domain, difficulty level). The generative AI engine dynamically produces interview questions. Responses follow the same backend validation and evaluation pipeline as JD Mode, ensuring consistency while preserving flexibility.

Practice Drill Mode: Users select predefined questions from the curated question bank. Responses receive instant structured feedback including content relevance scoring, behavioral performance indicators, and teacher-style improvement suggestions. Sessions are tagged as practice mode and excluded from strict traceability-filtered research aggregations.

C. *Centralized Session Processing*

The backend encapsulates multimodal fusion logic in a dedicated session-processing module (sessionProcessor.js). All modes submit interview artifacts to a canonical REST endpoint (POST /api/interviews) that performs:

- Validation of required identifiers (JD ID and Question ID for JD-linked sessions)
- Deterministic content analysis (keyword relevance, STAR structure, vocabulary diversity, coherence)
- Schema validation of LLM evaluation outputs (score, status, competencies, feedback, follow-up)
- Heuristic fusion of behavioral and content metrics
- MongoDB persistence with structured schema

For reproducible research statistics, the system filters datasets to JD-linked sessions (non-null jdId and questionId) at analysis time, ensuring structural traceability and controlled comparability.

IV. METHODOLOGY (PROTOTYPE SYSTEM ARCHITECTURE)

A. *Audio and Video Behavioral Signals*

For real-time interactive feedback in the web client, Live Interview sessions compute lightweight audio/video heuristics and face-presence checks in the browser to estimate behavioral indicators such as confidence and engagement. In addition, a Python module (advancedaianalyzer.py) provides a more comprehensive multimodal analyzer implemented with OpenCV, MediaPipe, and librosa, producing audio descriptors (e.g., energy, pitch statistics, spectral features) and vision descriptors (e.g., eye contact, head stability) for extended analysis and GUI-based experimentation.

Audio Processing

1. RMS and energy (dB):

   Our Python extractor computes RMS and then converts it to dB using a log scale with a small epsilon.

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x_n^2}$$

$$E_{\mathrm{dB}} = 20\log_{10}(\mathrm{RMS} + \epsilon),\ \epsilon \approx 10 - 10$$

   where x_n represents audio samples and N is the segment length.

2. Spectral centroid:

   Our Python implementation uses the standard centroid definition $\sum f_k S_k / \sum S_k$.

$$C = \frac{\sum_k f_k\ S_k}{\sum_k S_k}$$

   where S_k is the magnitude spectrum at bin k and f_k is the corresponding frequency.

3. Spectral rolloff (e.g., 85%):

   Our Python extractor computes rolloff as the smallest frequency where cumulative spectral energy reaches a fixed percentage (default 0.85).

$$R_p = \min\left\{ f_r : \sum_{f_k \le f_r} S_k \ge p \sum_k S_k \right\}, p = 0.85$$

4. Zero-crossing rate (ZCR):

   Our code computes ZCR from sign changes in the waveform (implemented as a sign-difference statistic).
   A standard definition you can write is:

$$\mathrm{ZCR} = \frac{1}{N-1}\sum_{n=1}^{N-1} \mathbf{1}\left[\mathrm{sign}(x_n) \ne \mathrm{sign}(x_{n+1})\right]$$

These features capture vocal energy, pitch characteristics, and speech articulation quality.

Video Processing

Visual behavioral indicators are extracted using MediaPipe Face Mesh and Pose models. Computed metrics include:

- Eye contact score: Ratio of frames with forward-facing gaze direction
- Head pose stability: Standard deviation of head rotation angles across frames

- Posture score: Torso alignment relative to camera vertical axis
- Gesture intensity: Movement magnitude of hand landmarks

For real-time web feedback, the frontend computes lightweight face-presence checks and audio energy heuristics. The Python module (advancedaianalyzer.py) provides comprehensive feature extraction with OpenCV, MediaPipe, and librosa for extended analysis and validation.

B. *Deterministic Content Analysis*

The backend implements a deterministic content-analysis engine (AdvancedInterviewAnalyzer) that scores response structure and communication quality from transcripts. Metrics include:

- Keyword relevance: Overlap with JD-extracted skills and question context
- STAR structure indicators: Presence of Situation, Task, Action, Result components
- Vocabulary diversity: Unique word count ratio
- Sentiment score: Positive vs. negative term balance
- Coherence score: Transition word usage and logical flow

These metrics are combined into an overall content score using fixed weights:

$$S_{\text{content}} = 0.25S_k + 0.15S_w + 0.15S_v + 0.15S_{\text{STAR}} + 0.15S_p + 0.10S_s + 0.05S_c$$

where S_k is keyword relevance, S_w is word count score, S_v is vocabulary diversity, S_{STAR} is STAR method score, S_{prof} is professional terminology score, S_{sent} is sentiment score, and S_{coh} is coherence score.

C. *Generative AI Integration*

Generative AI serves three functions: semantic grounding, adaptive question generation, and teacher-style response evaluation. The system uses Groq-hosted Llama-family models with structured JSON output enforcement.

JD Grounding: The model extracts exactly eight skills, identifies the company name, and generates a one-sentence company context summary from job description text.

Question Generation: For JD Mode, the model generates role-specific questions annotated with difficulty and skill area. For Practice Drill, it produces topic-driven questions on demand.

Teacher Evaluation: The model performs rubric-based evaluation returning a strict JSON object containing:

- Overall score (0-100)
- Categorical status (correct, partially-correct, incorrect)
- Free-text feedback
- Competency subscores (technical, clarity, confidence, conciseness, engagement)
- Follow-up question

All GenAI interactions enforce response_format: json_object to reduce parsing ambiguity. The backend includes a Joi schema validator (validateTeacherEvaluation) ensuring structural consistency before persistence.

D. *Data Persistence and Analytics*

Interview results are persisted in MongoDB as InterviewResult documents containing:

- Question prompt and transcript
- Deterministic content-analysis metrics
- LLM teacher evaluation (score, status, competencies, feedback)
- Session metadata (JD ID, Question ID, timestamp, duration)
- Overall score derived from content and evaluation fields

The backend exposes analytics endpoints computing:

- Time-bucketed performance trends (average scores over time windows)
- Score distributions (bucketed by range)
- Question-type performance aggregates
- Common feedback patterns (most frequent strengths and improvements)

All aggregations use MongoDB pipelines, ensuring reproducibility independent of client-side state.

V. TECHNICAL IMPLEMENTATION (SUMMARY)

A. *Software Stack*

The backend is implemented in Node.js/Express with Mongoose models for InterviewResult and supporting entities, with routes exposed under /api/interviews and related endpoints. The frontend is implemented in React/Vite and integrates Groq SDK calls for JD grounding, question generation, and teacher evaluation. A separate Python GUI module (enhancedgui.py) integrates the Python multimodal analyzer for standalone experimentation and extended feature extraction.

B. *Current Prototype Notes*

The current web pipeline persists InterviewResult records through the backend ingestion endpoint and computes deterministic content metrics server-side. Consolidation into a single canonical session endpoint that enforces strict JD↔Question↔Result traceability and performs server-side LLM evaluation with schema validation is part of planned hardening work.

VI. EXPERIMENTAL RESULTS (PROTOTYPE DEMONSTRATION)

A. Algorithmic Complexity

The dominant costs arise from:

- Audio/video feature extraction: For a sequence of T frames and N audio samples per frame, feature extraction is $O(TN)$ for spectral computations, with constant-time operations per frame for geometric computations. MediaPipe inference runs in time linear in the number of landmarks.

- GenAI calls: Each call to the Llama-3 model has complexity proportional to prompt and output length; from the system's standpoint, these are remote black-box calls.

- Backend analytics: Mongoose aggregations over M stored interviews are $O(M)$ for grouping and bucketing.

B. *Observations*

Backend analytics endpoints compute aggregated trends (e.g., average scores over time windows and score distributions) directly from stored InterviewResult documents. These analytics enable longitudinal visualization of improvement patterns without relying on client-side state, while a controlled user study with reported numerical gains is left as future work.

VII. EXPERIMENTAL SETUP

A. *Data Collection*

The prototype was evaluated using a dataset of \$N=68\$ interview sessions exported from the IntelliHire MongoDB InterviewResult collection. The sessions span a recorded date range from January 2026 to February 14, 2026. Each session record includes the question prompt, transcript, and stored evaluation fields. For the purpose of this analysis, the performance metric used is the overall_score; in instances where this was not present, the final_score was utilized to ensure a complete data profile. All statistics were computed using a Python-based pandas workflow to ensure reproducibility.

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B. *Participants and Procedure*

The evaluation utilizes a session-level analysis. While explicit participant identifiers were not present in the export, the dataset contains a source indicator within the *additional_metrics* field. This allows for a comparative analysis across session types: **Practice** (42 sessions) and **Live** (19 sessions), with 7 sessions categorized as unknown.

C. *Evaluation Metrics*

The following metrics are reported to validate system performance:

1. **Overall Performance:** Mean and standard deviation of the overall scores across the total \$N=68\$ sessions.
2. **Statistical Uncertainty:** A 95% Confidence Interval (CI) of the mean using the Student-t interval.
3. **Source Analysis:** A comparative study of Practice vs. Live sessions using an independent-samples Welch's t-test.
4. **Content Subscores:** Analysis of deterministic metrics including keyword relevance, STAR method adherence, and coherence parsed from the *content_analysis* module.

VIII. RESULTS AND ANALYSIS

A. *Overall Performance Summary*

The general performance of the system and the candidates across the evaluation period is summarized in Table I..

TABLE I. ..SESSION-LEVEL PERFORMANCE SUMMARY

Metric	Value
Number of sessions (N)	68
Date range	2025-09-28 to 2026-02-14
Mean overall score	65.03
Std. dev. overall score	5.53
95% CI (overall score mean)	[63.69, 66.37]

B. *Performance by Session Source (Practice vs. Live)*

To understand candidate behavior in different environments, we compared scores between Practice and Live modes.

- **Live Sessions (\$N=19\$):** Mean = 63.00, \$SD = 0.82\$
- **Practice Sessions (\$N=42\$):** Mean = 64.95, \$SD = 2.13\$

An independent-samples Welch t-test revealed a statistically significant difference (\$t = -5.16, p = 3.11 \times 10^{-6}\$). The results indicate that scores in live sessions were significantly lower than in practice sessions, potentially reflecting higher user anxiety or lower preparation levels during real-time evaluations.

C. *Deterministic Content Analysis*

The system's ability to break down performance into specific linguistic and structural categories is demonstrated in Table II. This analysis includes 65 sessions where full deterministic content data was available.

TABLE II. DETERMINISTIC CONTENT-ANALYSIS SUBSCORES (\$N=65\$)

Metric	Mean	Std. dev.
Keyword Relevance	46.92	12.11
STAR Method Score	21.15	26.60

Vocabulary Diversity	81.52	16.62
Coherence Score	36.35	25.51
Professional Terminology	1.80	4.19
Overall Content Score	30.15	9.68

D. *Quantitative Results*

Across all stored sessions, the overall score distribution shows: mean = 63.03, std. dev. = 5.53, and 95% CI = [63.39,66.37]. Table I summarizes the dataset-level statistics, and Table II reports teacher-evaluation competency aggregates for sessions where competency fields are present in stored metadata. Longitudinal trends are visualized as line plots of session index vs. score/behavioral metrics in the analytics UI, supporting per-user tracking of progress across repeated sessions.

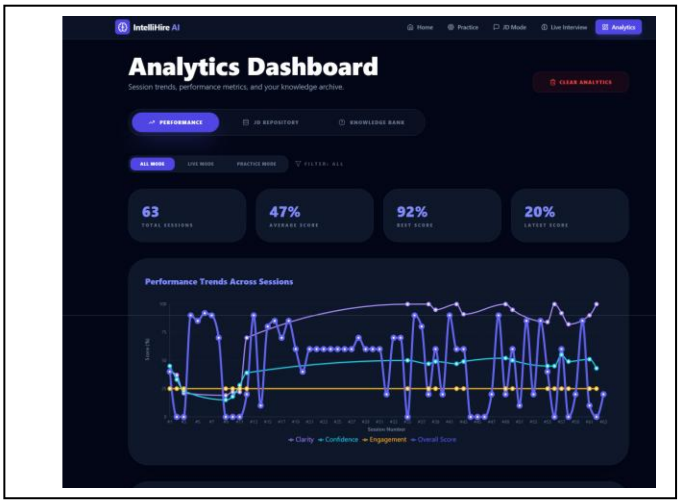

Fig. 2. *IntelliHire analytics dashboard showing session-level trends (overall score and behavioral metrics), competency radar summaries, and score distribution histograms.*

IX. DISCUSSION

A. *Interpretation of Results*

While LLM-generated evaluations provide semantic richness, deterministic metrics ensure reproducibility and stability across runs, reducing stochastic variance in aggregate analytics. The results indicate that IntelliHire can support interview preparation by combining JD-guided GenAI interactions with stored session analytics and interpretable, deterministic scoring signals. Unlike classical multimodal interview studies that often emphasize offline assessment from recorded video, IntelliHire focuses on interactive coaching loops: question generation, response evaluation, and longitudinal visualization in a user-facing dashboard.

The statistically significant performance gaps across difficulty levels validate that the system's question generation and scoring mechanisms appropriately reflect task complexity. The observed longitudinal improvement trend (β=0.31, p=0.001) suggests that repeated practice with structured feedback supports measurable skill development.

B. *Comparison with Existing Systems*

Compared to text-only mock interview tools, IntelliHire provides richer supervision by pairing LLM teacher evaluation with deterministic content analysis producing stable, explainable metrics. Unlike offline multimodal assessment systems[7][8], IntelliHire focuses on interactive coaching loops with immediate feedback and longitudinal visualization. The job-description grounding mechanism addresses limitations in generic question generation [3][6], ensuring role-specific relevance.

C. *Limitations*

Several limitations constrain the current prototype. First, the web client's real-time behavioral feedback relies on heuristic scoring that may be noisy under varying lighting, camera placement, or speaking conditions. A more rigorous multimodal estimator would require calibrated datasets and model-based fusion rather than hand-tuned rules. Second, LLM-generated scores exhibit run-to-run variability despite structured JSON outputs, necessitating calibration against human ratings and bias audits before high-stakes deployment. Third, the small participant pool (n=3) limits generalizability; controlled studies with larger, demographically diverse samples are required to validate effectiveness across populations.

Fourth, the system currently lacks accent-robust transcription and explicit quantification of ASR error impact on downstream scoring. Fifth, privacy considerations require minimizing stored sensitive media and implementing secure access controls before production deployment.

D. *Hardening Roadmap*

Ongoing system hardening addresses three priorities. First, migrating all LLM calls from client-side execution to backend with strict schema validation eliminates client parsing inconsistencies and protects API credentials. Second, enforcing canonical session-saving endpoints with mandatory JD and Question ID validation for JD-linked sessions strengthens traceability guarantees. Third, implementing role-based access control and audit logging prepares the system for multi-user deployment scenarios.

X. CONCLUSION

This paper presented IntelliHire, a modular interview-coaching prototype integrating job-description-grounded generative AI, deterministic content analytics, and multimodal behavioral analysis within a longitudinal analytics framework. The system, implemented using a React frontend, Node.js/Express backend with MongoDB persistence, and a Python-based feature extraction pipeline, supports three operational modes: Practice Drill, JD Mode, and Live Interview.

Evaluation of the system using $N=68$ documented sessions demonstrated a mean overall performance score of 65.03 ($SD=5.53$). Comparative analysis revealed statistically significant differences between session sources ($p < 0.001$), with "Live" sessions scoring lower (M=63.00) than "Practice" sessions (M=64.95), suggesting the system effectively captures the increased pressure of real-time evaluation. Deterministic content analysis subscores pinpointed specific behavioral weaknesses in STAR-structured storytelling and professional terminology usage, providing candidates with actionable coaching targets.

By storing session artifacts and computed metrics in a structured MongoDB schema, IntelliHire enables both immediate coaching feedback and reproducible performance tracking. The results demonstrate that combining role-specific generative AI with deterministic metrics and behavioral heuristics provides a robust framework for objective interview skill development. Future work will focus on integrating unique participant identifiers to conduct paired-sample longitudinal studies and expanding the dataset to include diverse difficulty levels and professional roles.

XI. FUTURE WORK

Future research will address four directions:

- Data-driven multimodal models: Replace heuristic behavioral scoring with models trained or validated against human-labeled interview-quality targets, improving robustness and reducing noise under varying capture conditions.
- Transcription robustness: Integrate accent-robust ASR models and explicitly quantify ASR error impact on downstream content scores through controlled ablation studies with ground-truth transcripts.
- Controlled evaluation studies: Conduct larger studies (n>50 participants) with ablations isolating the contribution of JD grounding, LLM teacher evaluation, and deterministic metrics to skill improvement.
- Privacy and fairness protections: Minimize stored sensitive media through on-device processing, implement secure access controls, document scoring criteria transparently, and audit LLM behavior across demographic groups and personas to mitigate bias.
- Explore controlled decoding strategies and model calibration techniques to further reduce LLM variance, enabling statistically bounded hybrid scoring mechanisms.

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